

The Measure of Advantage

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When Warren Buffet evaluates a company he looks for “sustainable competitive advantage.” Michael Porter, who stimulated so much excitement about this concept with his 1984 book, *Competitive Advantage*, said “the fundamental basis of above average performance in the long run is sustainable competitive advantage.” Today, in thousands of management programs and executive training sessions the cornerstone course is “strategy” and the central concept in these courses is “sustainable competitive advantage.”

The basic definition of “competitive advantage” is hardly rocket science. If your business can produce at lower cost than can competitors, or if it can deliver more perceived value than can competitors, then you have a competitive advantage. Since buyers differ in their locations, knowledge, tastes and other things, most advantages will only extend so far. Thus, Whole Foods had an advantage over Albertson’s only among grocery shoppers with good incomes who place a high value on organic and “natural” foods. Apple’s iPhone has an advantage among buyers who are willing to pay for the slickest Internet browsing experience on a cell phone. Among those who only want to text-message, the advantage disappears.

Defining “sustainability” is a bit more subtle. To see sustainability, you have to shift your focus from products and buyers to the resources that underlie an advantage. These resources are usually intangibles like brand names, reputations, relationships, special knowledge and skill accumulated over time, as well as patents, trademarks, and other

enforceable property rights. A competitive advantage is sustainable when the special resources supporting it are *scarce* in that competitors cannot get more of them. One cannot obtain an advantage from something that is not scarce.

Scarcity, either natural or contrived, grows out impediments to the creation of similar resources by competitors or new entrants. I have termed such impediments *isolating mechanisms*.¹ One basic isolating mechanism is the combination of *natural* scarcity and enforceable property rights. There is, for example, only so much beachfront property in Malibu and courts will enforce each owner's right to its exclusive use. Notice that an increase in demand for beachfront lots will make all Malibu beachfront property owners wealthier. Under this sort of natural scarcity, the owners don't really compete with one another—their competition is from less desirable property elsewhere.

One example of an isolating mechanism protecting a single owner is a patent. The patent gives its owner the legally enforceable right to monopolize the use of a technology for a time. Copyrights and brand names also create enforceable property rights that may protect the owner from direct duplication. More complex forms of isolating mechanisms include reputations, commercial and social relationships, network effects, dramatic economies of scale, distributed knowledge,² tacit knowledge and skill gained through experience, and causal ambiguity.

Some isolating mechanisms are attached to specific properties that can be individually bought and sold—things like patents, copyrights, and brand names. Others are more difficult to separate from the totality of the business—things like reputation and

¹ The phrase comes from biology where it describes any barrier to interbreeding. See Richard P. Rumelt, "Towards a Strategic Theory of the Firm," in R. Lamb (ed.), *Competitive Strategic Management*, Prentice-Hall, 1984, p. 556-70. Reprinted in N. J. Foss (ed.), *Resources, Firms, and Strategies*, Oxford University Press, 1998.

distributed knowledge. Whether separable or not, all are “resources” that draw value from scarcity and all are components of the total value of the business.

The Silver Machine

In 2002 my colleague Steven Lippman and I began to explore the connections between the idea of competitive advantage and various business performance measures in common use. We thought it would be straight forward, but we soon found ourselves mired in a swamp of fuzzy definitions and logical disconnects.

To help clear the air, we created a thought experiment—a very simplified model that stripped away most of the complexity present in real firms and real industries. This thought experiment led to the following exercise:

The silver-machine is a one-of-a-kind device left behind by a passing UFO. It disgorges 16.2 pounds of pure silver pellets every hour. It uses no inputs of energy or matter. It has an on-off switch and the inscription “Never runs down.” Given a silver market price of \$4.4 per ounce, the machine can produce \$10 million worth of silver each year, about 0.4% of world production. The constant market interest rate is 10% per year. There are no taxes. A wealthy buyer purchases the silver-machine from its original finder for \$100 million and sells the output at the market price. Does the buyer have a competitive advantage in the silver industry?

We purposely left out any quality advantage, any possibility of differentiation. Silver is a commodity and there are no niches, no market segments. What could be simpler? The machine makes silver at no cost. But confusion is only a step away. A long-established measure of business performance is return on invested capital (ROIC)— the ratio of annual profit to the amount of money that has, over time, been invested into the business.

The original discoverer of the silver-machine had invested nothing. Her ROIC seems to be infinite because the denominator of the ratio is zero. However, the wealthy buyer

paid \$100 million, so that his ROIC is $10/100 = 10\%$, the same as a bondholder's return (in this imaginary world). Was the competitive advantage destroyed by the change in ownership?

Steven and I could see that the new owner received only an ordinary return on investment. Yet we also felt that the silver-machine was a low-cost producer. To gain some traction on this issue, I distributed the silver-machine exercise to ten PhD experts in strategy.

There is an old saying: "If two experts agree, one of them is redundant." These ten experts were certainly not redundant. Five said the buyer would have a competitive advantage in the silver business and the other five said the buyer would not. The five experts who gave the edge to the silver-machine were impressed with its low cost of production. One wrote: "The most basic advantage is cost advantage, which the silver-machine definitely has. After all, zero production cost is as good as it gets."

Taking the other side of the issue, experts 6, 7, and 8 argued that the issue of advantage turned on the price paid for the machine. As one of them put it, "A competitive advantage allows a business to earn a higher-than average return on investment. Since the buyer paid \$100 million for the machine, the annual profit of \$10 million is a distinctly average 10% return." Expert 9 focused on the machine's market value rather than the price paid. Expert 9 wrote: "The economic profit is the cash profit of \$10 million less the opportunity cost of the capital tied-up in the machine. That opportunity cost is 10% of the \$100 million in capital tied up, or \$10 million. Thus, the economic profit is zero." Expert 10 argued that the silver-machine didn't really compete. "It just sits there and spits out silver like a bond pays interest," he wrote.

Steven and I had defined this machine so that it would have a clear differential advantage—as the first expert put it, "zero production cost is as good as it gets." Yet, one-half of the experts burdened the machine with various types of arguments which

apparently erased its advantage. It was disturbing to find this kind of disagreement among PhD strategy experts concerning a core concept in strategy.

The facile way of dealing with this confusion is to argue that real strategists create advantage; they don't argue over its technical definition. This might satisfy a textbook writer, but it falls short of the needs of academic strategic management.

Of the five respondents who denied any advantage to the purchased silver machine, one felt it was not a real "firm." That is, they argued that only a human organization could have a competitive advantage. It remains to be determined how that person would assess the division of a company that owned the silver machine.

Three argued that the return on investment was normal; thus, there was no advantage. If you define advantage as return on investment, this is logical. But this may be a bit constrained. I recall consulting with a small division of Bethlehem Steel in 1978. It used a specialized rolling mill designed to make the wide, flat automobile hoods popular in the earlier Lincoln Continentals. The manager's problem was that changed styles had shrunk demand for his mill's capacity so that his ROI was low- unacceptably low by corporate standards. What should he do? The mill had no other obvious use. It ran at about one-half capacity, generating a reasonable cash flow. It was able to out-bid competitors for customers who still wanted its wide, flat, capability. By many standards, it had a competitive advantage in its niche. It was even "making money," though not as much as those who had approved the original investment had hoped. So, is competitive advantage the same thing as return on investment?

The fifth respondent simply said that the machine did not really "compete." That is, it did not plan, price, market, or even have any employees. It just sat there spitting out silver. I am sympathetic to this view, but this is a thought experiment. If you accept the UFO premise, perhaps you should stretch your definition of a firm.